

Chapter 5

Developing Budget

A project's budget is a function of the project's task or activities, the duration of tasks and activities, their sequence, and the resources. Generally, resources deployed on a project will have a cost. So the cost of using a particular task/activity must be taken into account in overall project budget.

Until these costs are accounted for, the project manager and the organisation will not know the true cost of the project.

Cost Estimation

Estimating the cost of a particular activity or task with an estimated duration involves 5 steps:

- ✓ Defining what resources will be needed to carry out the work
- ✓ Determining the quantity of resources that are needed
- ✓ Defining the cost of using each resource
- ✓ Calculating the cost of the task or activity
- ✓ Ensuring that the resources are levelled i.e. resources have not been over allocated

For example let us assume that we have identified a specific task and estimated that it will take 1 day to complete plus requires 1 project team member. Let us also assume, for simplicity, that no other resources are needed. This estimate may need that we define a cost for using this particular resource e.g. if our team member earns 2000 frs an hour, that sum is what our employee sees on his/her pay check (before taxes and other deductions).

The organisation however, may provide certain benefits to the employee i.e. health care, life insurance, and so on that should be included in the cost for using this particular resource. Since these costs are going to vary from organisation to organisation, let us assume that your friends in the accounting department have conducted a cost accounting analysis for us plus the real cost of using this particular employee i.e. hourly wage plus benefits is 2000 frs an hour.

Consequently, if we pay our employee for 1 day's work (i.e. an eight-hour day), then the cost of completing this particular task is:

Cost of task = Estimated duration x True cost of the resource

$$= 8 \text{ hours} \times 2000 \text{ frs}$$

$$= 16000 \text{ frs}$$

We can equally estimate the cost of a salaried employee by prorating his salary. This simply means that we can assign a portion of that salary to the task at hand e.g. let us take that the fully loaded or true annual cost to the organisation is 200,000 frs.

If this employee works a five-day work week, the associated true cost to the organisation would be for $5 \times 52 = 260$ days a year. Therefore, prorated cost per day would be $200,000/260$ work days = 769.23 frs.

Nevertheless, this whole process can be greatly simplified if use a project management software tool. We still have to identify the tasks and accurately estimate their durations. So determining the cost of a particular task and for the entire project becomes effortless. Figure 7.9 (Marchewka, 2003) shows how resources can be assigned to specific tasks on a project.

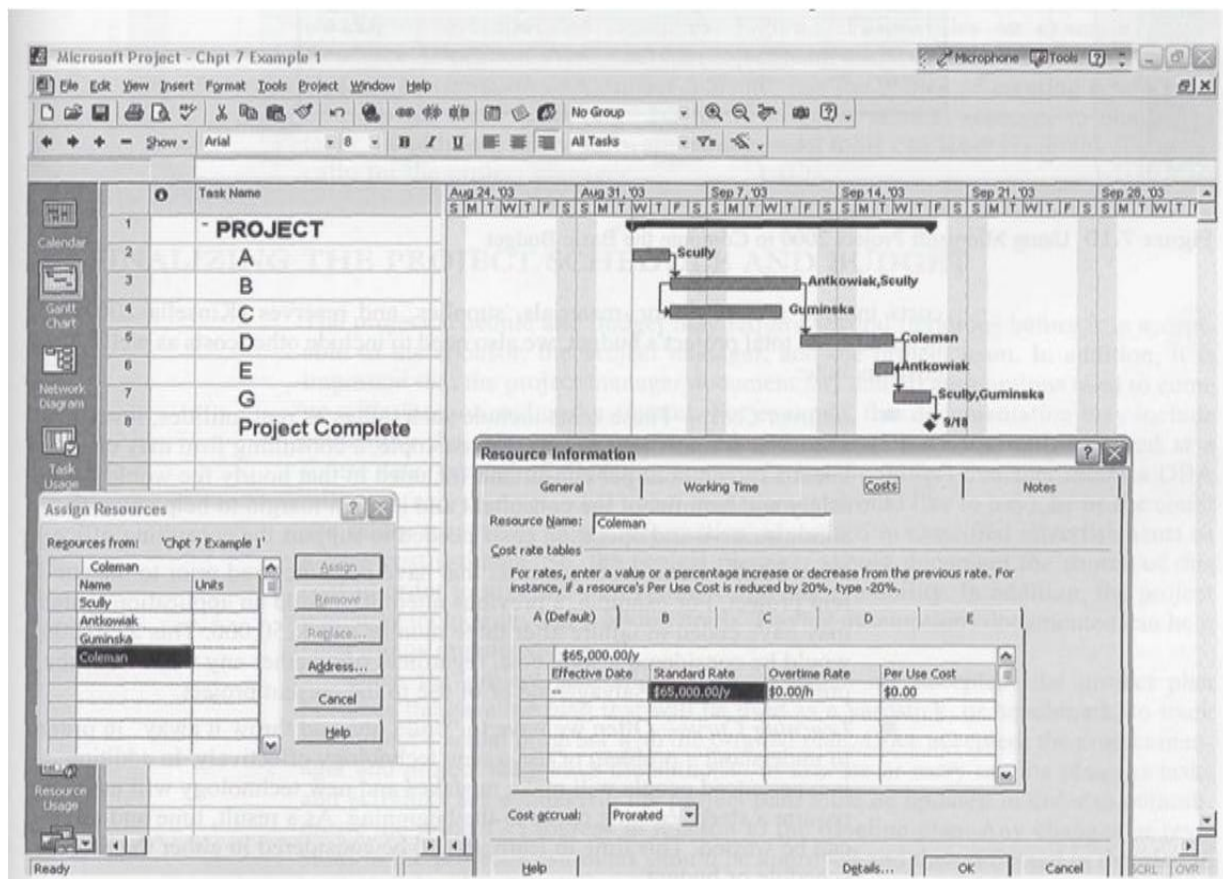


Figure 7.9 Using Microsoft Project to Assign Resources to Tasks

A project's total budget is computed using a bottom-up approach. By summing individual costs for each task/activity. As depicted in Figure 7.10 (Marchewka, 2003), the basic budget for this project is \$5,203.85.

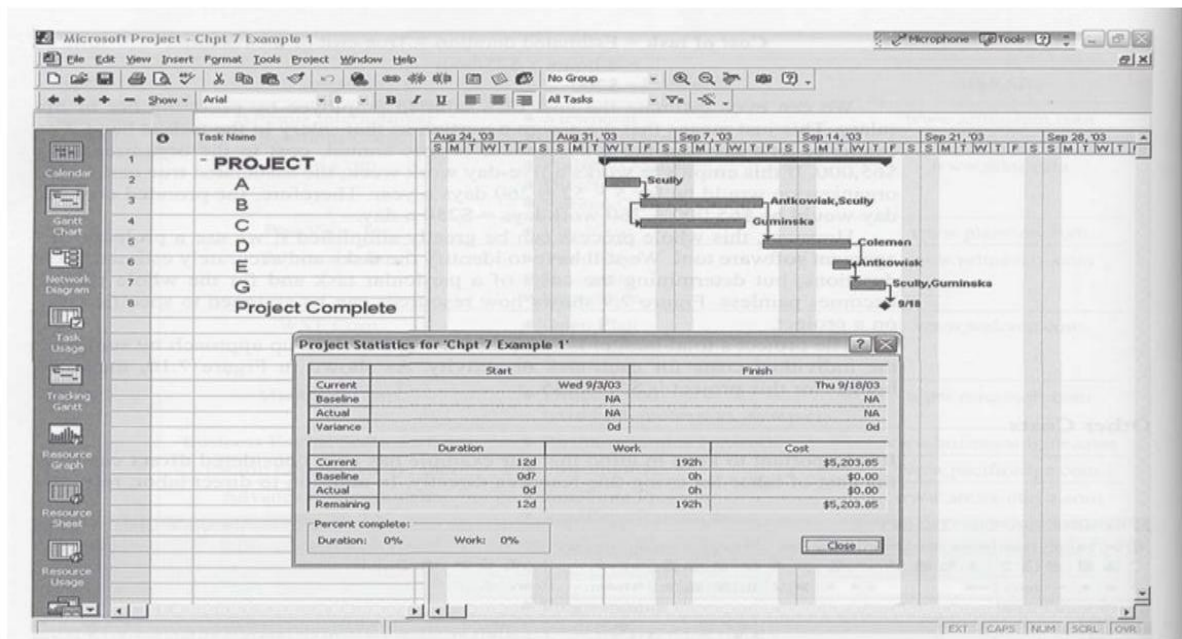


Figure 7.10 Using Microsoft Project 2000 to Compute the Basic Budget
Marchewka, 2003

Other Costs 1/8

It is worth keeping in mind that our example has only considered direct costs or cost of labour for using the resource directly. In addition to direct labour, resource costs include direct labour, materials, supplies and reserves (Kinsella, 2002). In determining the total project's budget, we also need to include other costs as well. These include:

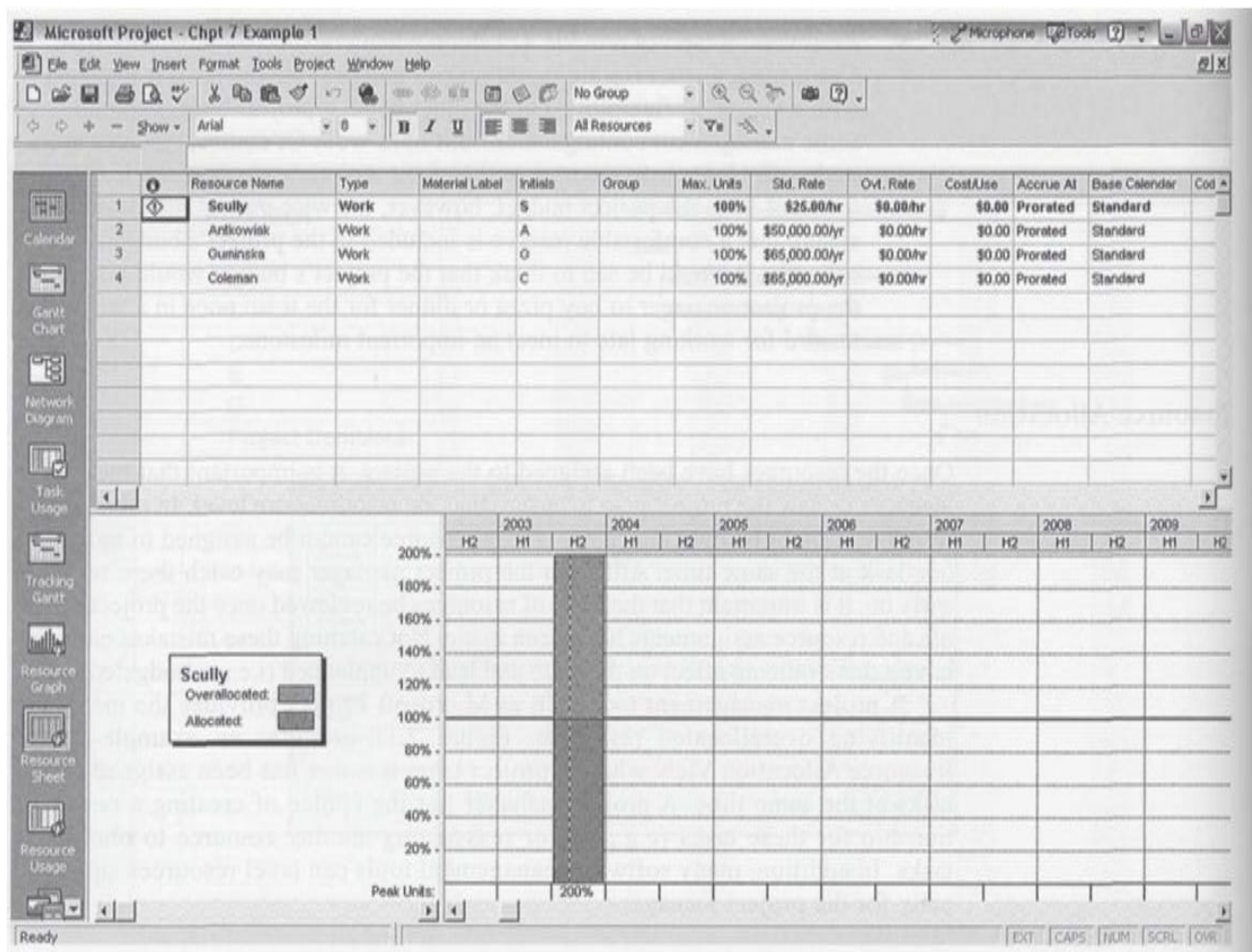
- ☐ **Indirect Costs:** they include such things as rents, utilities, insurance and other administration costs e.g. a consulting firm may charge a client \$150 an hour per consultant. Included in that hourly fee could be the salary, and benefits of the consultant plus enough margin to help cover administration and operation costs required to support consulting office.
- ☐ **Sunk Costs:** these are costs that have been incurred prior to current project e.g. a previous attempt to build an app system may have ended in failure after 3 months and \$250000. This \$250000 would be considered as a sunk cost irrespective of whether any work from previous project is salvageable or of use to the current project.
- ☐ **Learning Curve:** often we have to “build one and throw it away” in order to understand a problem, or a new technology effectively. In addition, inexperienced people will make mistakes and new technology will normally require a steep learning curve in the beginning. As such, time and effort will be wasted. This time to learn should be factored in either the project schedule or budget.
- ☐ **Reserves:** they provide a cushion when unexpected situations arise. Contingency reserves are based on risk and provide project manager with a degree of flexibility. On the other hand, a project budget shall have some management contingencies built in as well. Normally, reserves are a trade-off. Upper management or client will regard these as fat that

can be trimmed from budget. However, a clever project manager will ensure that a comfortable reserve is included in the project's budget e.g. it would be sad to think that project's budget would not allow project manager to pay for pizza or dinner for team once in a while as a reward for working late to meet a milestone.

Resource Allocation

Once resources have been assigned to the project, it is vital that project manager review project plan to ensure resources are level. In other words, resources cannot be over allocated i.e. a resource cannot be assigned to more than one task at the same time. Even though project manager may catch these mistakes early on, it is vital that the levels of resources be reviewed once project schedule and resource assignments have been made.

Not catching these mistakes early can have a demoralising effect on team plus lead to unplanned (i.e. unbudgeted) costs. Project management tool such as MS Project provides a means for identifying over allocated resources. Figure 7.11 (Marchewka) provides an example of Resource Allocation View.



Where a project team member has been assigned to two tasks at the same time, project manager has the choice of creating a new relationship for these tasks or reassigning another resource to one of the tasks. In addition, many software management tools can level resources automatically for project manager.

IT project Quality Management

What is Quality?

Before answering that, bear in mind that quality can mean different things to different people e.g. if we were comparing the quality of two cars – an expensive luxury car with leather seats and every possible option to a lower-priced economy car which gets you basically to where you want to go. Several people may be compelled to say that the more expensive car has a higher quality. Even though the more expensive car has more features, one may not regard it a bargain if you have to keep bringing it back to the shop for expensive repairs. Thus, the less-expensive car may start looking much better to you if it were more dependable or met higher safety standards.

On the other hand, why do car manufacturers produce different models of cars with different price range? If everyone could afford luxury cars, then quality comparison between different manufacturers would be much easier. Despite the fact that you may have your eyes on luxury cars, your current financial situation may be a constraint. As such, you would have to buy a car you can afford. Therefore, it's vital to define quality only in terms of features or functionality. Other attributes such as dependability or safety may be just as crucial to customer. Similarly in software development, we build systems that have a great deal of functionality but perform poorly. On the other hand, we can create systems that have few features or limited functionality, but also fewer defects.

The dictionary defines quality as “an inherent or distinguishing characteristic; a property, or as something having a high degree of excellence.” In business, quality has been defined in terms of “fitness for use” and “conformance to requirements.” “Fitness for use” focuses on delivering a system that meets customer's needs while “conformance to requirements” centres more on meeting some predefined set of standards. Thus quality depends on the needs or expectations of the customer. It's up to project manager/team to accurately define those needs or expectations while permitting customer to remain within his/her resource constraints.

Although concepts and philosophies of quality have received a great deal of attention over the last 50 years in manufacturing and service sectors, several of these ideas have been integrated into a relatively new discipline or knowledge area called project quality management (PQM).

Project management body of knowledge defines PQM as: the process required to ensure that the project will satisfy the needs for which it was undertaken. It includes all activities of the overall management function that determine the quality policy, objective and responsibility and implements them by means of quality planning, quality assurance, quality control and quality improvement within the quality system (1995).

Furthermore, PMBOK defines major quality management processes as:

- ❖ Quality Planning – establishes which quality standards are vital to the project plus deciding how these standards will be met.
- ❖ Quality Assurance – evaluates overall project performance frequently to ensure that project team is meeting the specified quality standards

- ❖ Quality Control – monitoring the activities and results of project to ensure that the project conforms with the quality standards

In addition, project organisation as a whole should deploy this information to eliminate the causes of unsatisfactory performance and implement new processes and techniques to improve project quality throughout the project organisation.

Thus, PQM should focus on both the product and the process of the project. From our own perspective, the projects most important is the information system (IS) that the project team must deliver. The system must be “fit for use” and “conform to specified requirements” outlined in both project’s scope and requirements definition.

More crucially, IT product must add measurable value to sponsoring organisation and meet the scope, schedule and budget objectives. Quality can nevertheless, equally be built into project management and software development processes. A process refers to activities, methods, materials, and measurements used to produce the product or service. By focusing on both the product & chain of project processes, project organisation can use its resources More efficiently and effectively, minimising errors and meet or exceed project stakeholder expectations.

The cost of quality, however, can be viewed as the cost of conforming to standards (i.e. building quality into the product and processes) as well as the cost of not conforming to the standards (i.e. rework). Substandard levels of quality can be viewed as waste, errors, or the failure to meet the project sponsor’s or client’s needs, expectation, or system requirements (Kloppenberg & Petrick, 2002). Failure to meet quality requirements or standards can have negative consequences for all project stakeholders and impact the other objectives. To be more specific, adding additional work or repeating project activities will mostly extend the project schedule plus expand the project budget.

Thus, poor quality can be an embarrassment for project manager, project team and project organisation e.g. one of the most widely publicised software defect stories was the faulty baggage-handling software at Denver International Airport. Bugs in the software delayed the opening of the airport from October 1993 to February 1995 at the cost of \$1,000,000 a day!! Newspaper accounts reported that bags were literally chewed up and contents of bags were flying through the air (Williamson, 1997).

The concepts and philosophies of quality management have received a great deal of attention over the years. Despite being popularised by Japanese, several organisations in different countries have initiated quality management program. Such programs include: ISO certification, six steps to sigma initiatives or awards such as Deming Prize with Malcolm Baldrige National Qaulity Award. More recently, the Capability Maturity Model (CCM) has provided a framework for software quality which focuses on assessing the process maturity of software development within an organisation.

PQM as depicted in Figure 10.1 will not only include the concepts, teaching, tools and methods of quality management but also validation and verification control.

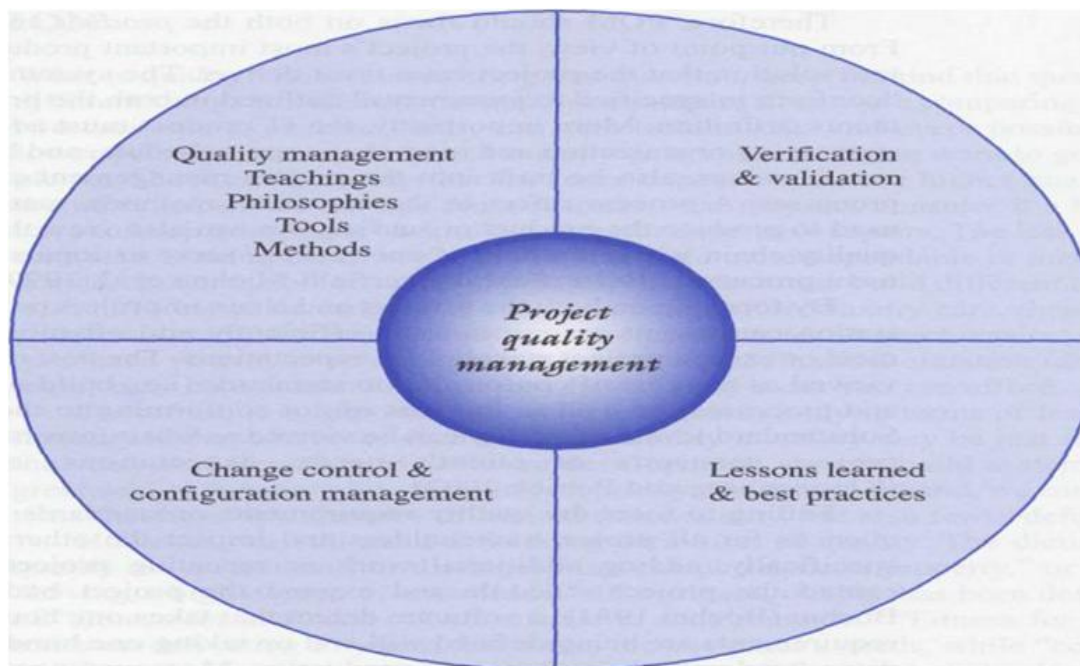


Figure 10.1 Project Quality Management

Verification and validation (V&V) activities within PQM should be performed throughout project life cycle. They need project team to continually ask:

- Are we building the right product?
- Are we building the product the right way?

Thus, project quality plan should not only focus on final testing of system at the end of the project life cycle, but also on all project deliverables. Finding and fixing problems earlier in the project life cycle is less costly than when having to address them in the later stages of the project. Finding problems does not only lead to less rework later, but also save project manager and project team from having to deal with embarrassing issues and problems since the project's product is in the hands of sponsors or end-customer.

Also, software development frequently requires a number of people to work on multi-versions of documents, programs and database files that are shared and distributed among various project stakeholders. Change control in the form of configuration management thus is a method of code and document management to track and organise different versions of documents and files. It keeps project team more focused plus minimised the likelihood of errors. Knowledge management (KM) and lessons learned equally can be implemented as best practices plus incorporated in projects throughout the organisation. The outcome of such changes is both continuous improvement and a maturing of IT project management process.

Concepts of quality management, V and V activities, change control and KM support the overall PQM plan. The plan does not only aid in improving the overall quality of project's product and processes but can also result in a competitive advantage for the organisation since project will have a greater likelihood of accomplishing its expected organisational value and support the scope, schedule and budget objectives.

Monitor and Control

Quality control pays attention on monitoring the activities and project results to make sure that project complies with the quality standards. Once project's standards are in place, it's vital to monitor them to certify that project quality objective is accomplished.

Equally, control is crucial for identifying problems in order to make improvements once a process is under control. Similar to quality assurance activities, quality control should be ongoing throughout the life cycle of the project and only end when the customer or sponsor accepts the final IT solution (Kloppenborg & Petrick, 2002). Furthermore, quality control embodies monitoring and controlling activities with respect to the product, processes and project deploying system concepts as illustrated in Figure 10.10.

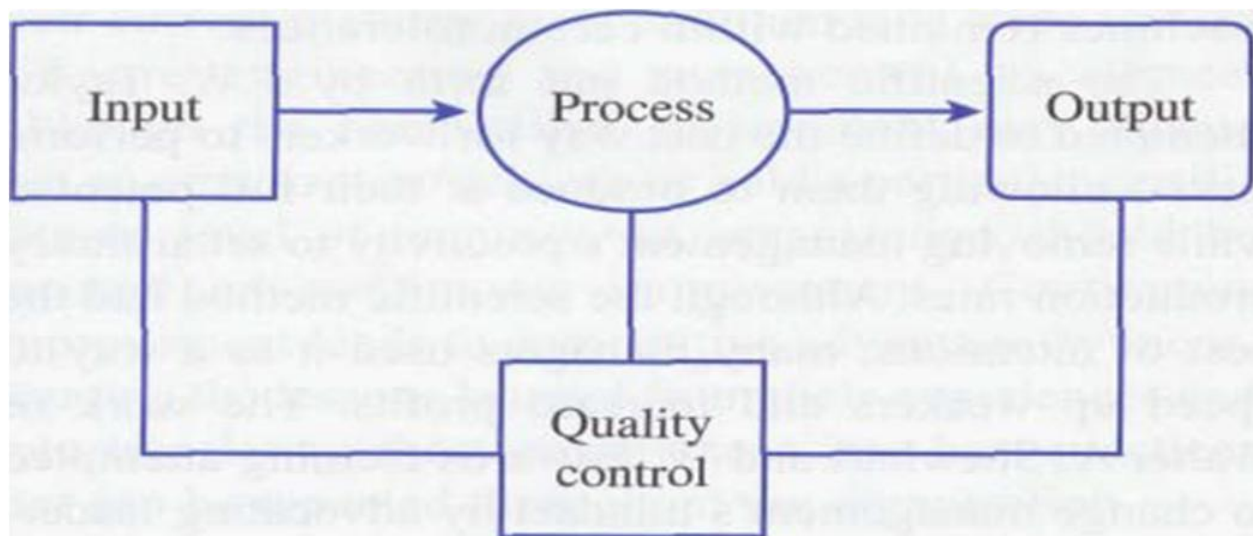


Figure 10.10 Quality Control Activities, Marchewka, 2003

Quality control activities must focus on the inputs and outputs of each process. If inputs to a process are of poor quality, then output of a particular process will be of poor quality too since as a rule, a process may not be capable of changing inherent quality of the input. Also, even if the input to a process is of high quality, the process itself may create an output with lower quality or product if the requirements are not properly defined.